

**ТРИГЕРЪТ КАТО ЕЛЕМЕНТАРЕН
СТРУКТУРЕН АВТОМАТ.
ВИДОВЕ ТРИГЕРИ.
СИНТЕЗ НА ТРИГЕРИ**

ТРИГЕРЪТ КАТО ЕЛЕМЕНТАРЕН СТРУКТУРЕН АВТОМАТ

Елемент с две устойчиви състояния

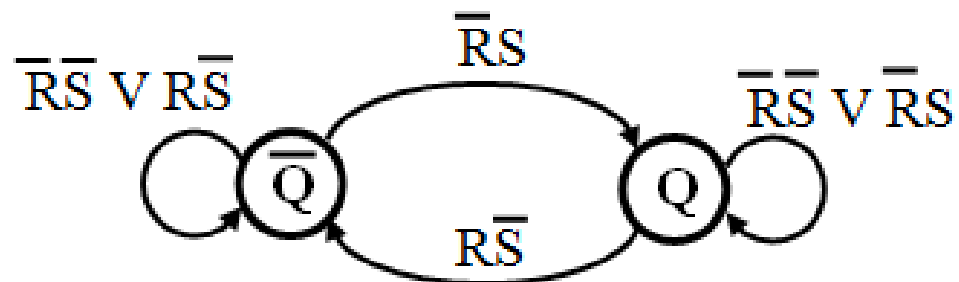
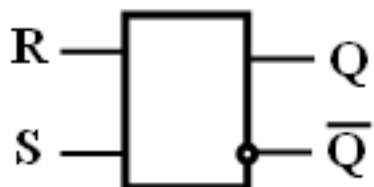
Определение:

Двоичен елемент, който може да се превключва в кое да е от **две възможни състояния** при определени условия и който *съхранява последното си състояние* (до постъпване на друго управляващо въздействие) *в продължение на нужния интервал от време*, се нарича **тригер**.

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Изходният сигнал на тригера се променя **само при промяна на неговото състояние**, т. е. той функционира като **автомат на Мур** (елементарен автомат на Мур).

RS-ТРИГЕР



R	S	Q^t	Q^{t+1}
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	*
1	1	1	*

R	S	Q^{t+1}
0	0	Q^t
0	1	1
1	0	0
1	1	*

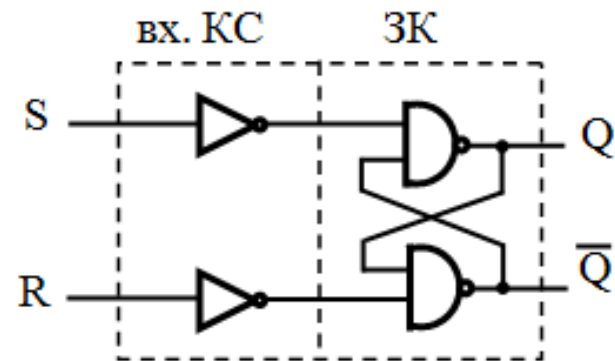
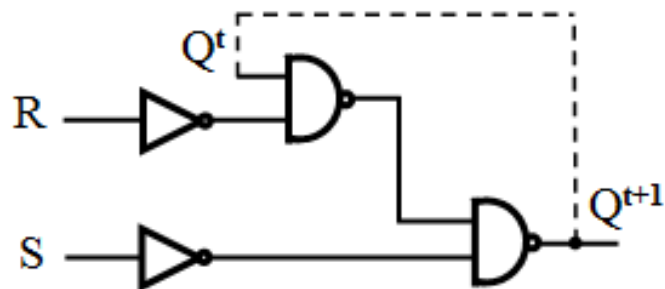


RS-ТРИГГЕР

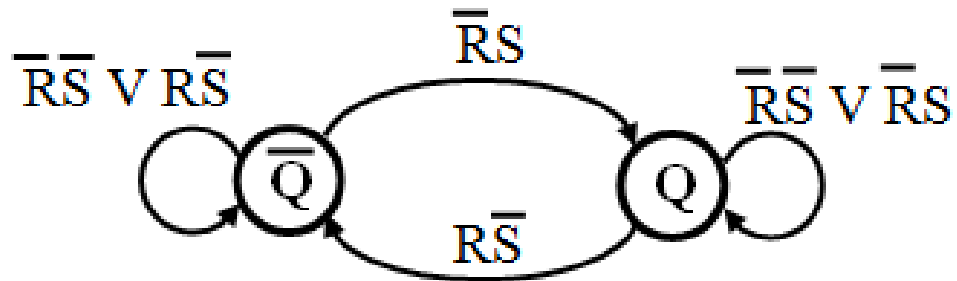
	SQ			
R	00	01	11	10
0	0	1	1	1
1	0	0	*	*

$$Q^{t+1} = \bar{R}.Q^t + S$$

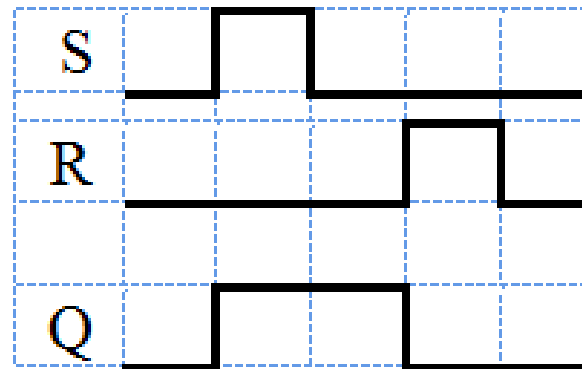
$$Q^{t+1} = \overline{\bar{R}.Q^t.\bar{S}}$$



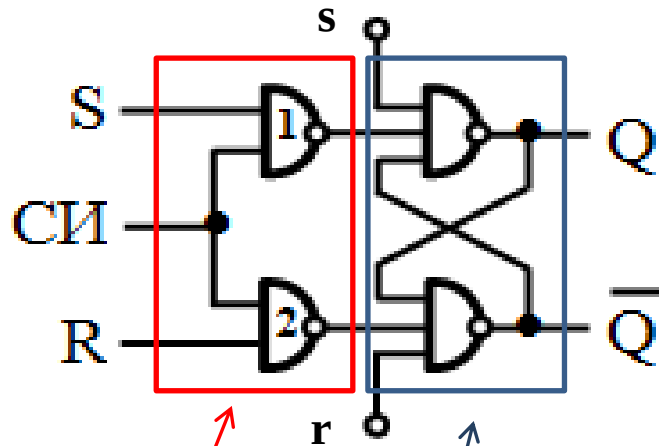
RS-ТРИГЕР



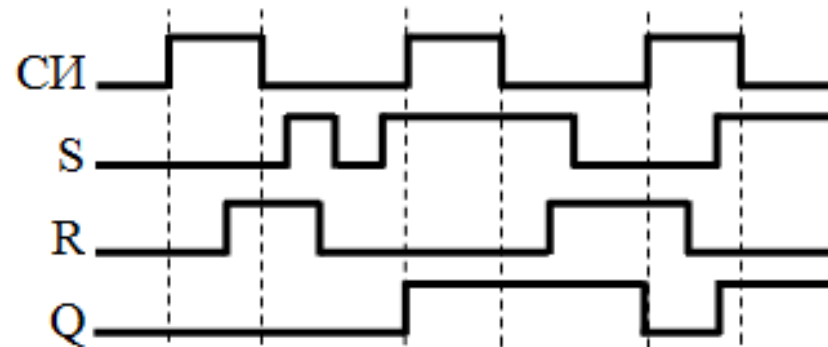
R	S	Q^{t+1}
0	0	Q^t
0	1	1
1	0	0
1	1	*



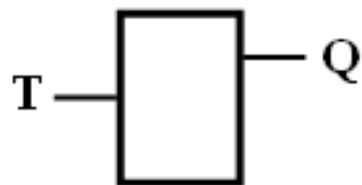
RS-ТРИГЕР



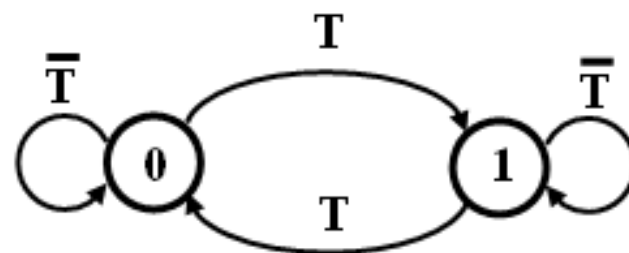
$\bar{R}$	$\bar{S}$	R	S	Q^t	Q^{t+1}		$\bar{R}$	$\bar{S}$	R	S	Q^{t+1}
1	1	0	0	0	0		1	1	0	0	Q^t
1	1	0	0	1	1		1	0	0	1	1
1	0	0	1	0	1		0	1	1	0	0
1	0	0	1	1	1		0	0	1	1	*
0	1	1	0	0	0						
0	1	1	0	1	0						
0	0	1	1	0	*						
0	0	1	1	1	*						



T-ТРИГГЕР



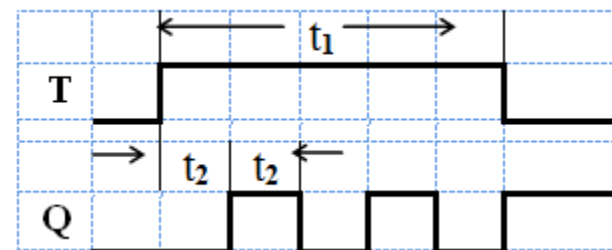
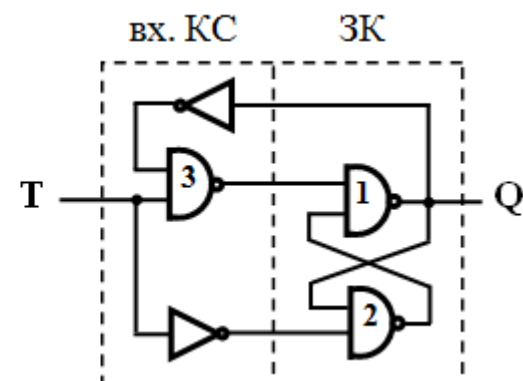
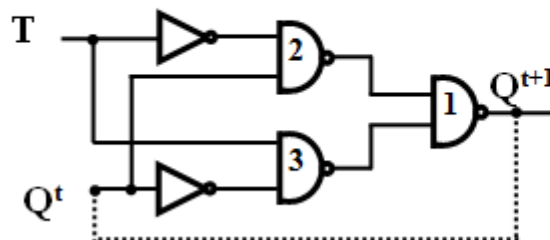
T	Q^t	Q^{t+1}
0	0	0
0	1	1
1	0	1
1	1	0



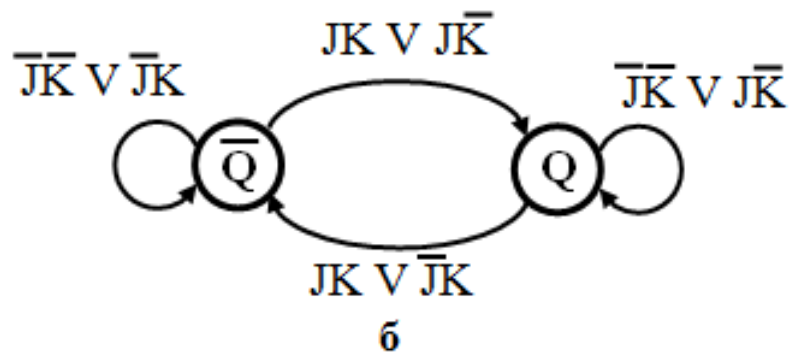
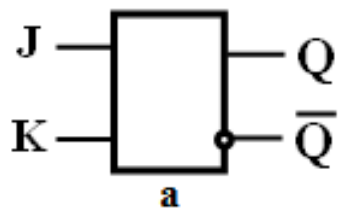
T	Q^t	Q^{t+1}
0	0	1
1	1	0

$$Q^{t+1} = (\bar{T}Q^t + T\bar{Q}^t)$$

$$Q^{t+1} = \overline{\overline{\overline{T}} \cdot \overline{\overline{Q^t}} \cdot \overline{\overline{T}} \cdot \overline{\overline{Q^t}}}$$



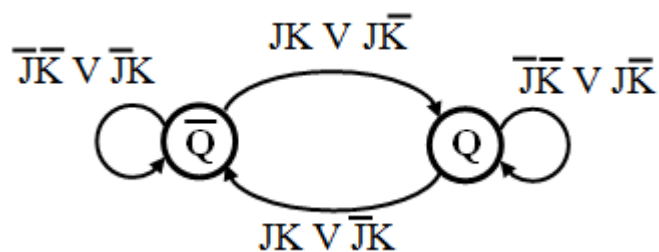
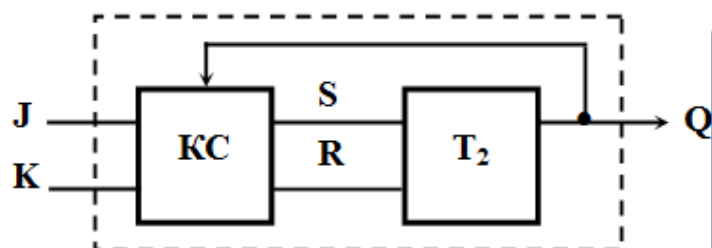
JK-ТРИГГЕР



J	K	Q^t	Q^{t+1}	R	S	Q^{t+1}
0	0	0	0	0	0	Q^t
0	0	1	1	0	1	1
0	1	0	0	1	0	0
0	1	1	0	1	1	*
1	0	0	1	г		
1	0	1	1			
1	1	0	1			
1	1	1	0			

в

JK-ТРИГГЕР

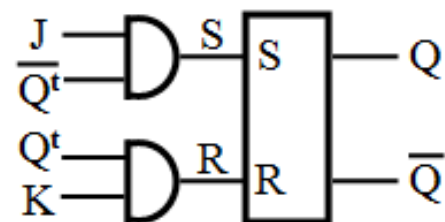


J	K	Q^t	Q^{t+1}	R	S
0	0	0	→ 0	*	0
0	0	1	→ 1	0	*
0	1	0	0	*	0
0	1	1	0	1	0
1	0	0	1	0	1
1	0	1	1	0	*
1	1	0	1	0	1
1	1	1	0	1	0

Q^t	Q^{t+1}	R	S
0 → 0		*	0
0	1	0	1
1	0	1	0
1 → 1		0	*

JK-ТРИГГЕР

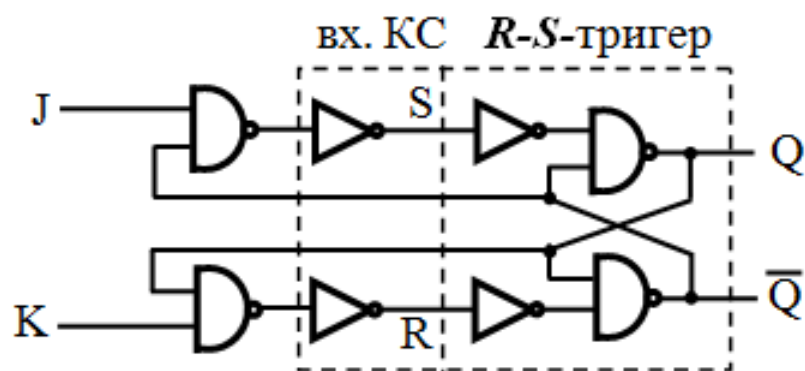
J	K	Q^t	Q^{t+1}	R	S		Q^t	Q^{t+1}	R	S
0	0	0	0	*	0		0	0	*	0
0	0	1	1	0	*		0	1	0	1
0	1	0	0	*	0		1	0	1	0
0	1	1	0	1	0		1	1	0	*
1	0	0	1	0	1					
1	0	1	1	0	*					
1	1	0	1	0	1					
1	1	1	0	1	0					



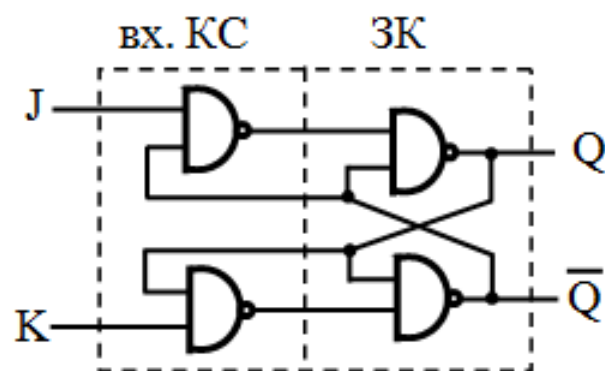
KQ					KQ				
J	00	01	11	10	J	00	01	11	10
0	*	0	1	*	0	0	*	0	0
1	0	0	1	0	1	1	*	0	1

$$R = K \cdot Q^t$$

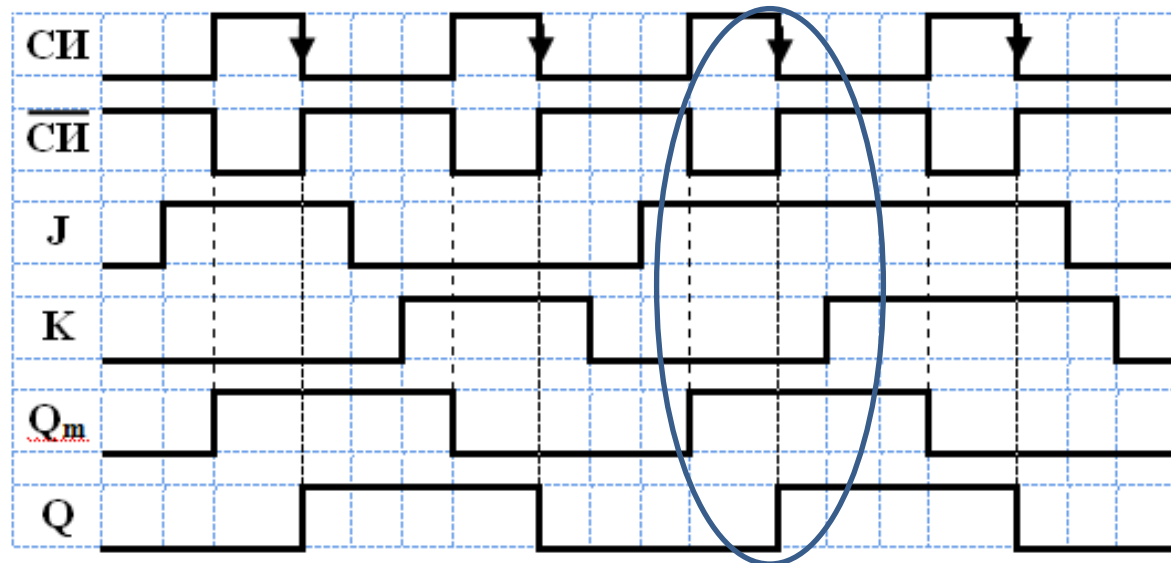
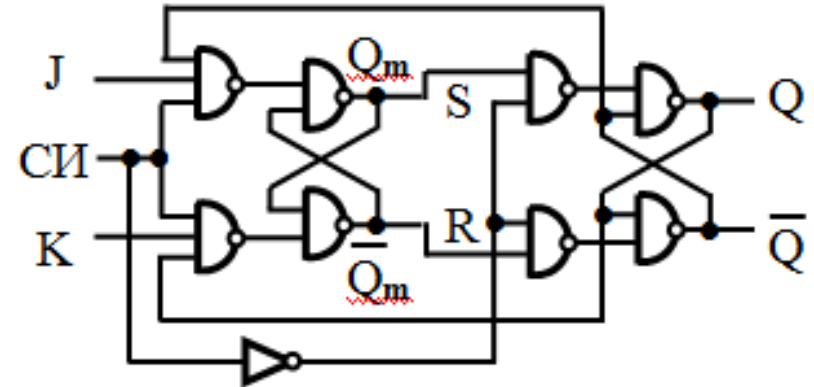
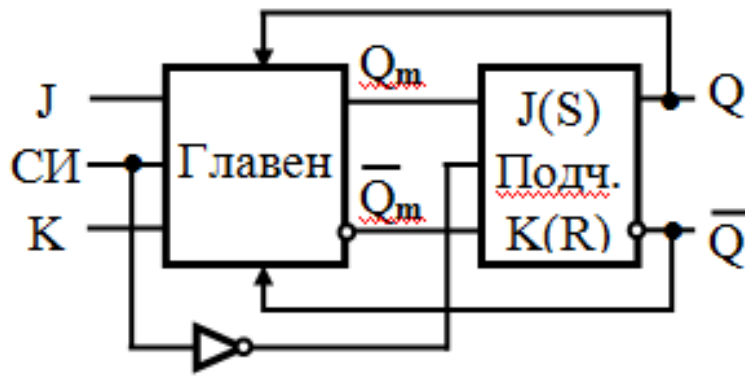
$$S = J \cdot \overline{Q^t}$$



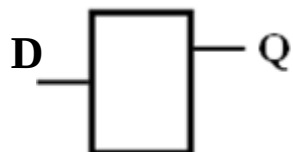
или



ТРИГЕРИ СЪС СТРУКТУРА “ГЛАВЕН-ПОДЧИНЕН”



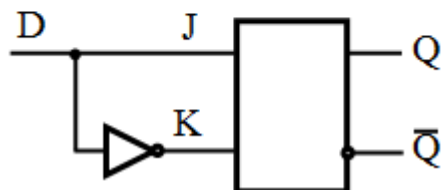
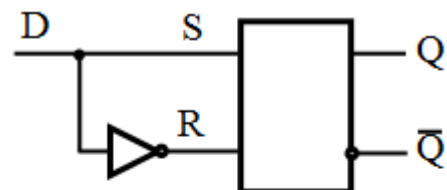
D-ТРИГЕР



$\underline{D}^t$	Q^t	Q^{t+1}
0	0	0
0	1	0
1	0	1
1	1	1

$$Q^{i+1} = D^i \cdot \overline{Q}^i + D^i \cdot Q^i = D^i$$

$$J = \overline{K} = D; S = \overline{R} = D$$



RS-; JK-, T-; D-ТРИГЕРИ

$Q^t \rightarrow Q^{t+1}$	R	S
0 → 0	*	0
0 → 1	0	1
1 → 0	1	0
1 → 1	0	*

$Q^t \rightarrow Q^{t+1}$	J	K
0 → 0	0	*
0 → 1	1	*
1 → 0	*	1
1 → 1	*	0

$Q^t \rightarrow Q^{t+1}$	D
0 → 0	0
0 → 1	1
1 → 0	0
1 → 1	1

$Q^t \rightarrow Q^{t+1}$	T
0 → 0	0
0 → 1	1
1 → 0	1
1 → 1	0

ОБОБЩЕНИЕ

Класификация на тригерите

- Според закона на функциониране – **RS-, JK-, D-, T- тригери**
- Според начина на запис на информацията в тригера – **синхронни и асинхронни**
 - синхронни по ниво;
 - синхронни по фронт с вътрешна задръжка;
 - синхронни по фронт със структура “главен - подчинен”.